Answer key: Lesson 3, Digit Cards

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. You have two digit cards: 3 and 5. Use both cards to make two numbers.

Answer: 35 and 53. 53 is bigger.

With 3 on the left the number is 3 tens and 5 ones: 35. With 5 on the left it is 5 tens and 3 ones: 53. The same two cards make two numbers. 53 is bigger, because 5 tens is more than 3 tens.

Watch for: Writing 8 (3 + 5).; Writing 35 twice.

S2. Write the bigger number each time.

Answer: 72, 48, 65, 30

(a) 7 tens is more than 2 tens: 72. (b) Same tens, 8 ones is more than 1 one: 48. (c) 6 tens is more than 5 tens: 65. (d) 3 tens is more than 1 ten: 30.

Watch for: Choosing the number with the bigger ones digit (saying 13 is bigger than 30).; Not looking at the tens first.

S3. Mei's bars and squares are in the picture. Ben has 4 tens bars and 1 ones square.

Answer: Mei 38, Ben 41, Ben has more

Mei has 3 bars and 8 squares: 38. Ben has 4 bars and 1 square: 41. 4 tens is more than 3 tens, so Ben has the bigger number, even though Mei has more squares.

Watch for: Saying Mei has more because she has 8 squares.; Reading Mei's picture as 83.

S4. You have the cards 6 and 2. Make the biggest and the smallest number.

Answer: 62 and 26

The biggest card goes on the left for the biggest number: 62. The smallest card goes on the left for the smallest number: 26.

Watch for: Writing 62 for both.; Writing 8 (6 + 2).

S5. Write these numbers in order. Start with the smallest: 52, 25, 41, 14.

Answer: 14, 25, 41, 52

14 has 1 ten, 25 has 2 tens, 41 has 4 tens, 52 has 5 tens. Smallest first: 14, 25, 41, 52.

Watch for: Putting 25 after 41 because 5 is bigger than 1.; Writing them biggest first.

S6. Ravi says 19 is bigger than 21, because 9 is bigger than 2. Is he right? Write why.

Answer: No. 21 is bigger. 2 tens is more than 1 ten.

19 is 1 ten and 9 ones. 21 is 2 tens and 1 one. 2 tens is more than 1 ten, so 21 is bigger. Ravi looked at the ones digits first. Look at the tens first.

Watch for: Saying Ravi is right because 9 is more than 2.; Not giving a reason (any clear sentence about the tens is fine).

S7. You have the cards 2 and 8. Use both cards each time.

Answer: 82 and 28

82 has 8 tens, which is more than 5 tens. 28 has only 2 tens, which is less than 5 tens.

Watch for: Writing 82 for both parts.; Writing 8 and 2.

S8. Siti has 25 stickers. Ravi has 52. Draw bars and squares. Who has more?

Answer: Ravi (52 is 5 tens and 2 ones; 25 is 2 tens and 5 ones)

Siti: 2 bars and 5 squares. Ravi: 5 bars and 2 squares. 5 bars is more than 2 bars, so Ravi has more stickers, even though both numbers use a 2 and a 5.

Watch for: Saying they have the same because they use the same digits.; Drawing 2 and 5 single squares for Siti.

Think harder (page 3)

T1. You have three digit cards: 3, 5 and 8. Use any two cards. What is the biggest number you can make? What is the smallest?

Answer: 85 and 35

Numbers you can make: 35, 38, 53, 58, 83, 85. The biggest is 85 (8 tens, the biggest card on the left, then the next biggest card). The smallest is 35.

Watch for: Using all three cards.; Writing 88 (using the 8 twice).

T2. You have three digit cards: 0, 3 and 7. Use any two cards to make a two-digit number. What is the smallest you can make? What is the biggest? Be careful with the 0.

Answer: 30 and 73

The 0 can not go on the left (03 is just 3). The two-digit numbers are 30, 37, 70 and 73. The smallest is 30 and the biggest is 73.

Watch for: Writing 03 or 3 for the smallest.; Writing 37 for the smallest.

T3. Cards 1, 3, 5 and 7. Make two-digit numbers with two different cards. Clue 1: the number is more than 40. Clue 2: the left digit is bigger than the right digit. Write every number that fits.

Answer: 51, 53, 71, 73, 75

More than 40 means a tens card of 5 or 7. With 5 on the left: 51, 53, 57. With 7 on the left: 71, 73, 75. Clue 2 removes 57, because 5 is not bigger than 7. Left: 51, 53, 71, 73, 75.

Watch for: Stopping after one or two numbers.; Writing 57 (5 is less than 7).

T4. You have the cards 2, 4 and 6. How many different two-digit numbers can you make with two different cards? Write them all.

Answer: 6 numbers: 24, 26, 42, 46, 62, 64

2 on the left: 24, 26. 4 on the left: 42, 46. 6 on the left: 62, 64. That is 2 + 2 + 2 = 6 numbers.

Watch for: Missing some by listing in a random order.; Writing 22, 44 or 66 (each card can be used only once).

Go further (page 4)

G1. In a 3-digit number like 357, the left card counts the hundreds. You have the cards 4, 0 and 9. Use all three cards. Write the biggest and the smallest number. The 0 can not go on the left.

Answer: 940 and 409

The ways are 409, 490, 904, 940 (049 and 094 start with 0, so they are not 3-digit numbers). The biggest is 940. The smallest is 409.

Watch for: Writing 049 as the smallest.; Writing 904 as the biggest (the 9 is on the left, but 4 should come next).

G2. You have the cards 1, 2, 3 and 4. A two-digit number uses two cards. How many different numbers can you make if each card is used only once? And if you may use the same card twice, like 11?

Answer: 12 and 16

Each card once: 4 choices for the tens card, then 3 choices for the ones card, so 4 x 3 = 12. A card may be used twice: 4 choices and 4 choices, so 16 (the 12 and the four numbers 11, 22, 33, 44).

Watch for: Counting only 8 or 6 numbers.; Forgetting 11, 22, 33 and 44 in part b.

G3. Use the cards 2, 5 and 8 once each to make a 3-digit number. Clue 1: more than 500. Clue 2: the ones digit is bigger than the tens digit. Clue 3: less than 600. What number am I?

Answer: 528

The numbers are 258, 285, 528, 582, 825, 852. More than 500: 528, 582, 825, 852. Less than 600: 528, 582. Ones digit bigger than tens digit: 528 (8 is bigger than 2). 582 has 2 and 8 the wrong way round.

Watch for: Stopping at 582 and not checking clue 2.; Listing only some of the 6 numbers.

Investigation (page 5): Is the bigger card always first?

For two different cards, the bigger card on the left always makes the bigger number, because the tens decide. With the cards 0 and 8 you can only make 80, because 08 is just 8. With the same card twice (4 and 4) you can only make 44. So you can make two different two-digit numbers only when the two cards are different and neither is 0.

Look for: A child who tests several pairs, notices that the tens digit decides, and finds the two special cases (a 0 card, two matching cards). A reason using tens and ones is better than "because it is bigger".

Exit ticket (page 5)

E1. Write the bigger number: 39 or 41.

Answer: 41

41 has 4 tens and 39 has only 3 tens, so 41 is bigger. Look at the tens first.

Watch for: Writing 39 because 9 is bigger than 1.

E2. You have the cards 7 and 3.

Answer: 73 and 37

The biggest card on the left makes the biggest number: 73. The smallest card on the left makes the smallest number: 37.

Watch for: Writing 73 for both.; Writing 10 (7 + 3).

E3. Write these numbers in order, smallest first: 63, 36, 60.

Answer: 36, 60, 63

36 has 3 tens. 60 and 63 both have 6 tens. 60 has 0 ones and 63 has 3 ones, so 60 comes before 63.

Watch for: Putting 60 before 36 or after 63.; Writing them biggest first.

Sneak-peek puzzles (not in the workbook)

1. 18 stars.
2. Mei. Her number is 84, which is 8 tens. Ali has 48, which is only 4 tens.